

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Trigonometric Power Integral

Evaluate the definite integral:

$$\int_0^{\frac{\pi}{3}} \frac{1}{\sin^6(x) + \cos^6(x)} dx$$

Solution: Let I be the given integral. We can rewrite the integrand by factoring out $\cos^6(x)$:

$$I = \int_0^{\frac{\pi}{3}} \frac{\sec^6(x)}{\tan^6(x) + 1} dx = \int_0^{\frac{\pi}{3}} \frac{\sec^4(x) \sec^2(x)}{\tan^6(x) + 1} dx$$

Substitute $u = \tan(x)$, which gives $du = \sec^2(x) dx$. The limits of integration change from $x = 0 \implies u = 0$ and $x = \pi/3 \implies u = \sqrt{3}$. Since $\sec^2(x) = 1 + u^2$, we have $\sec^4(x) = (1 + u^2)^2$:

$$I = \int_0^{\sqrt{3}} \frac{(1 + u^2)^2}{u^6 + 1} du$$

Notice that the denominator can be factored as a sum of cubes: $u^6 + 1 = (u^2)^3 + 1^3 = (u^2 + 1)(u^4 - u^2 + 1)$. Substitute this back into the integral:

$$I = \int_0^{\sqrt{3}} \frac{(1 + u^2)^2}{(u^2 + 1)(u^4 - u^2 + 1)} du = \int_0^{\sqrt{3}} \frac{u^2 + 1}{u^4 - u^2 + 1} du$$

Divide the numerator and denominator by u^2 :

$$I = \int_0^{\sqrt{3}} \frac{1 + \frac{1}{u^2}}{u^2 - 1 + \frac{1}{u^2}} du$$

Let $w = u - \frac{1}{u}$. Then $dw = \left(1 + \frac{1}{u^2}\right) du$. Also note that $w^2 = u^2 - 2 + \frac{1}{u^2} \implies u^2 + \frac{1}{u^2} = w^2 + 2$. The denominator becomes $w^2 + 2 - 1 = w^2 + 1$. As $u \rightarrow 0^+$, $w \rightarrow -\infty$. As $u \rightarrow \sqrt{3}$, $w \rightarrow \sqrt{3} - \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$.

$$I = \int_{-\infty}^{\frac{2}{\sqrt{3}}} \frac{1}{w^2 + 1} dw = [\arctan(w)]_{-\infty}^{\frac{2}{\sqrt{3}}}$$

$$I = \arctan\left(\frac{2}{\sqrt{3}}\right) - \left(-\frac{\pi}{2}\right) = \arctan\left(\frac{2\sqrt{3}}{3}\right) + \frac{\pi}{2}$$

1.2 Hard Integral

Logarithmic Polynomial Integral

Evaluate the definite integral:

$$\int_0^1 x^2 \ln(1-x) dx$$

Solution: Let's use the substitution $u = 1 - x$, which implies $x = 1 - u$ and $dx = -du$. The limits of integration change from $x = 0 \Rightarrow u = 1$ to $x = 1 \Rightarrow u = 0$.

$$I = \int_1^0 (1-u)^2 \ln(u) (-du) = \int_0^1 (1-u)^2 \ln(u) du$$

Expanding the polynomial:

$$I = \int_0^1 (1 - 2u + u^2) \ln(u) du = \int_0^1 \ln(u) du - 2 \int_0^1 u \ln(u) du + \int_0^1 u^2 \ln(u) du$$

We use the standard identity for integrals of this form: $\int_0^1 u^n \ln(u) du = -\frac{1}{(n+1)^2}$ for $n > -1$. Evaluating each term using this identity: 1) For $n = 0$: $\int_0^1 \ln(u) du = -\frac{1}{(0+1)^2} = -1$ 2) For $n = 1$: $-2 \int_0^1 u \ln(u) du = -2 \left(-\frac{1}{(1+1)^2} \right) = -2 \left(-\frac{1}{4} \right) = \frac{1}{2}$ 3) For $n = 2$: $\int_0^1 u^2 \ln(u) du = -\frac{1}{(2+1)^2} = -\frac{1}{9}$ Summing the results:

$$I = -1 + \frac{1}{2} - \frac{1}{9} = -\frac{18}{18} + \frac{9}{18} - \frac{2}{18} = -\frac{11}{18}$$

Differentials

2.1 Medium Differential

Limit of a Linear Second-Order ODE

Let $y(x)$ satisfy the equation:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2$$

Find the value of $w_2 = \lim_{x \rightarrow -\infty} 2y(x)$.

Solution: This is a second-order linear non-homogeneous differential equation with constant coefficients. First, we find the general solution to the corresponding homogeneous equation $y'' - 3y' + 2y = 0$. The characteristic equation is:

$$r^2 - 3r + 2 = 0 \implies (r - 1)(r - 2) = 0$$

The roots are $r = 1$ and $r = 2$. The homogeneous solution is:

$$y_h(x) = C_1e^x + C_2e^{2x}$$

Next, we find a particular solution $y_p(x)$ for the non-homogeneous equation. Since the right side is a constant (2), we guess a constant solution $y_p = A$. Substituting y_p into the differential equation gives:

$$0 - 3(0) + 2A = 2 \implies 2A = 2 \implies A = 1$$

The general solution to the differential equation is:

$$y(x) = y_h(x) + y_p(x) = C_1e^x + C_2e^{2x} + 1$$

Now we evaluate the required limit as $x \rightarrow -\infty$:

$$\lim_{x \rightarrow -\infty} y(x) = \lim_{x \rightarrow -\infty} (C_1e^x + C_2e^{2x} + 1)$$

Since $e^x \rightarrow 0$ and $e^{2x} \rightarrow 0$ as $x \rightarrow -\infty$, the first two terms vanish regardless of the initial conditions (the constants C_1 and C_2).

$$\lim_{x \rightarrow -\infty} y(x) = 1$$

Therefore, the value of w_2 is:

$$w_2 = 2 \left(\lim_{x \rightarrow -\infty} y(x) \right) = 2(1) = 2$$

2.2 Hard Differential

Generalized Operator Differential Equation

Let $y(x; n)$ satisfy the generalized equation $L_n y = n$, where L_n is a differential operator defined as:

$$L_n = \prod_{j=1}^n (D - j)$$

Knowing that the sequence w_n is defined as $w_n = \lim_{x \rightarrow -\infty} 2^n y(x; n)$, find $\lim_{n \rightarrow \infty} w_n$.

Solution: The differential equation is an n -th order linear ODE: $\prod_{j=1}^n (D - j)y = n$. The characteristic equation for the homogeneous part is $\prod_{j=1}^n (r - j) = 0$, which has positive real roots $r = 1, 2, \dots, n$. Thus, the homogeneous solution $y_h(x)$ is a linear combination of exponential functions e^{jx} for $j \in \{1, \dots, n\}$. As $x \rightarrow -\infty$, every term in $y_h(x)$ approaches zero since $j > 0$. To find the limit, we only need the particular solution y_p . Since the right-hand side is a constant n , we guess a constant solution $y_p = A$. Applying the operator L_n to the constant A (where $DA = 0$):

$$L_n A = \prod_{j=1}^n (0 - j)A = A \prod_{j=1}^n (-j) = A(-1)^n n!$$

We equate this to the right-hand side of the differential equation:

$$A(-1)^n n! = n \implies A = \frac{n}{(-1)^n n!} = \frac{(-1)^n}{(n-1)!}$$

Since $y(x; n) = y_h(x) + A$ and $y_h(x) \rightarrow 0$ as $x \rightarrow -\infty$, we have:

$$\lim_{x \rightarrow -\infty} y(x; n) = \frac{(-1)^n}{(n-1)!}$$

Now we find the sequence w_n :

$$w_n = 2^n \lim_{x \rightarrow -\infty} y(x; n) = 2^n \frac{(-1)^n}{(n-1)!} = \frac{(-2)^n}{(n-1)!}$$

Finally, we compute the limit of w_n as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} w_n = \lim_{n \rightarrow \infty} \frac{(-2)^n}{(n-1)!}$$

Since factorials grow infinitely faster than exponential functions, the sequence converges to 0.

$$\lim_{n \rightarrow \infty} w_n = 0$$

Derivatives

3.1 Medium Derivative

Tetration Chain Rule

Let $f(x) = x^{x^x}$. Find $f'(0.5)$.

Solution: To differentiate $f(x) = x^{x^x}$, we use logarithmic differentiation. Take the natural logarithm of both sides:

$$\ln(f(x)) = x^x \ln(x)$$

Differentiate both sides with respect to x using the product rule:

$$\frac{f'(x)}{f(x)} = \frac{d}{dx}(x^x) \cdot \ln(x) + x^x \cdot \frac{d}{dx}(\ln(x))$$

To find the derivative of x^x , we use logarithmic differentiation again: let $g(x) = x^x$, so $\ln(g(x)) = x \ln(x)$.

$$\frac{g'(x)}{g(x)} = \ln(x) + 1 \implies g'(x) = x^x(\ln(x) + 1)$$

Substitute this back into our primary equation:

$$\frac{f'(x)}{f(x)} = x^x(\ln(x) + 1) \ln(x) + x^x \left(\frac{1}{x} \right) = x^x(\ln^2(x) + \ln(x)) + x^{x-1}$$

Multiplying by $f(x) = x^{x^x}$:

$$f'(x) = x^{x^x} \left(x^x(\ln^2(x) + \ln(x)) + x^{x-1} \right)$$

Now evaluate at $x = 0.5 = 1/2$. Note that $(1/2)^{1/2} = \frac{1}{\sqrt{2}}$.

$$f(0.5) = (0.5)^{(0.5)^{0.5}} = \left(\frac{1}{2} \right)^{\frac{1}{\sqrt{2}}}$$

The terms inside the bracket:

$$x^x(\ln^2(x) + \ln(x)) = \frac{1}{\sqrt{2}} \left(\ln^2(0.5) + \ln(0.5) \right)$$

$$x^{x-1} = (0.5)^{-0.5} = \sqrt{2}$$

Combining them gives the exact value:

$$f'(0.5) = \left(\frac{1}{2} \right)^{\frac{1}{\sqrt{2}}} \left(\frac{\ln^2(0.5) + \ln(0.5)}{\sqrt{2}} + \sqrt{2} \right)$$

3.2 Hard Derivative

Exponential Function Product Rule

Let $f(x) = x^{x^3} e^{-x^3}$, $x > 0$. Find $f''(1)$.

Solution: We use logarithmic differentiation. Take the natural logarithm of both sides:

$$\ln(f(x)) = \ln(x^{x^3} e^{-x^3}) = x^3 \ln(x) - x^3$$

Differentiate with respect to x :

$$\frac{f'(x)}{f(x)} = \frac{d}{dx}(x^3 \ln(x) - x^3) = (3x^2 \ln(x) + x^3 \cdot \frac{1}{x}) - 3x^2$$

$$\frac{f'(x)}{f(x)} = 3x^2 \ln(x) + x^2 - 3x^2 = 3x^2 \ln(x) - 2x^2 = x^2(3 \ln(x) - 2)$$

So the first derivative is:

$$f'(x) = f(x)x^2(3 \ln(x) - 2)$$

Evaluate $f(1)$ and $f'(1)$:

$$f(1) = 1^{1^3} e^{-1^3} = 1 \cdot e^{-1} = \frac{1}{e}$$

$$f'(1) = f(1) \cdot 1^2 \cdot (3 \ln(1) - 2) = \frac{1}{e}(0 - 2) = -\frac{2}{e}$$

Now we differentiate $f'(x)$ to find the second derivative using the product rule on three terms ($f(x)$, x^2 , and $3 \ln(x) - 2$):

$$f''(x) = f'(x)[x^2(3 \ln(x) - 2)] + f(x) \frac{d}{dx}[x^2(3 \ln(x) - 2)]$$

$$f''(x) = f'(x)[x^2(3 \ln(x) - 2)] + f(x) \left[2x(3 \ln(x) - 2) + x^2 \left(\frac{3}{x} \right) \right]$$

$$f''(x) = f'(x)x^2(3 \ln(x) - 2) + f(x)[2x(3 \ln(x) - 2) + 3x]$$

Evaluate $f''(x)$ at $x = 1$:

$$f''(1) = f'(1) \cdot 1^2 \cdot (3 \ln(1) - 2) + f(1)[2(1)(3 \ln(1) - 2) + 3(1)]$$

Substitute the known values $f(1) = 1/e$ and $f'(1) = -2/e$:

$$f''(1) = \left(-\frac{2}{e} \right) (1)(-2) + \left(\frac{1}{e} \right) [2(-2) + 3]$$

$$f''(1) = \frac{4}{e} + \frac{1}{e}[-4 + 3] = \frac{4}{e} - \frac{1}{e} = \frac{3}{e}$$

Physics & Engineering

4.1 Mechanics

Sphere Rolling in a Hollow Cylinder

A uniform solid sphere of mass m and radius R is placed inside a fixed, rough, hollow cylinder of radius $3R$ whose longitudinal axis is oriented horizontally. The sphere is released from rest at an angular position $\theta_0 = \frac{\pi}{3}$ measured from the lowest point. Find the minimum coefficient of static friction $\mu_{\min}$ required for the sphere to roll without slipping immediately after release.

Solution: Let θ be the angle from the lowest point. The forces acting on the sphere are gravity (mg), the normal force (N), and static friction (f). Since we are evaluating the condition *immediately* after release, the velocity of the sphere is $v = 0$. Therefore, there is no centripetal acceleration, meaning the net radial force is zero:

$$N - mg \cos(\theta_0) = \frac{mv^2}{2R} = 0 \implies N = mg \cos(\theta_0)$$

The equation of motion along the tangential direction is:

$$mg \sin(\theta_0) - f = ma$$

Where a is the linear tangential acceleration of the sphere's center of mass. The rotational equation of motion about the sphere's center of mass is:

$$fR = I\alpha$$

For a uniform solid sphere, the moment of inertia is $I = \frac{2}{5}mR^2$. Since the sphere rolls without slipping, $a = \alpha R$.

$$fR = \left(\frac{2}{5}mR^2\right) \left(\frac{a}{R}\right) \implies f = \frac{2}{5}ma \implies a = \frac{5f}{2m}$$

Substitute this into the tangential force equation:

$$mg \sin(\theta_0) - f = m \left(\frac{5f}{2m}\right) = \frac{5}{2}f$$

$$mg \sin(\theta_0) = \frac{7}{2}f \implies f = \frac{2}{7}mg \sin(\theta_0)$$

To prevent slipping, the required static friction must satisfy $f \leq \mu_s N$. Therefore, the minimum coefficient $\mu_{\min}$ is:

$$\mu_{\min} = \frac{f}{N} = \frac{\frac{2}{7}mg \sin(\theta_0)}{mg \cos(\theta_0)} = \frac{2}{7} \tan(\theta_0)$$

Evaluate at $\theta_0 = \frac{\pi}{3}$:

$$\mu_{\min} = \frac{2}{7} \tan\left(\frac{\pi}{3}\right) = \frac{2\sqrt{3}}{7}$$

4.2 Electric Circuits

Resistor Power and Unknown Resistance

An ideal 18 V battery supplies a $4\ \Omega$ resistor in series with a parallel combination of a $12\ \Omega$ resistor and an unknown resistor R . The $12\ \Omega$ resistor dissipates 3 W. Find R in Ω , giving your answer to two decimal places.

Solution: First, we find the voltage across the $12\ \Omega$ resistor. The power P dissipated by a resistor is given by $P = \frac{V^2}{R}$.

$$3 = \frac{V_p^2}{12} \implies V_p^2 = 36 \implies V_p = 6\text{ V}$$

This voltage $V_p = 6\text{ V}$ is the voltage across the entire parallel combination (both the $12\ \Omega$ resistor and the unknown resistor R). By Kirchhoff's Voltage Law, the sum of voltage drops in the circuit must equal the battery voltage.

$$V_{\text{battery}} = V_{\text{series}} + V_p$$

$$18 = V_{\text{series}} + 6 \implies V_{\text{series}} = 12\text{ V}$$

The series resistor has a resistance of $4\ \Omega$. We can find the total current I_{total} flowing out of the battery and through the series resistor using Ohm's law:

$$I_{\text{total}} = \frac{V_{\text{series}}}{4} = \frac{12}{4} = 3\text{ A}$$

This total current splits when it reaches the parallel combination. We find the current I_{12} flowing through the $12\ \Omega$ resistor:

$$I_{12} = \frac{V_p}{12} = \frac{6}{12} = 0.5\text{ A}$$

By Kirchhoff's Current Law, the remaining current must flow through the unknown resistor R :

$$I_R = I_{\text{total}} - I_{12} = 3 - 0.5 = 2.5\text{ A}$$

Now we use Ohm's law to find the resistance R , knowing the voltage across it is $V_p = 6\text{ V}$ and the current is $I_R = 2.5\text{ A}$:

$$R = \frac{V_p}{I_R} = \frac{6}{2.5} = 2.4\ \Omega$$

Giving the answer to two decimal places, $R = 2.40\ \Omega$.

4.3 Quantum Mechanics

Photoelectric Current and Distance

A small metal surface is uniformly illuminated by light from a point source. At a distance $r_1 = 3.0\text{ m}$ the photoelectric current is $I_1 = 7.50\ \mu\text{A}$. The surface is moved to a distance $r_2 = 5.0\text{ m}$ with its orientation and illuminated area

unchanged. The photon energy remains above the work function. Find the new photoelectric current I_2 in μA .

Solution: Assuming the photon energy remains unchanged and above the work function, the photoelectric current (number of emitted electrons per second) is directly proportional to the intensity of the incident light (number of incident photons per unit area per second). For a point source emitting uniformly in all directions, the intensity of light obeys an inverse-square law with respect to the distance r :

$$\text{Intensity} \propto \frac{1}{r^2}$$

Therefore, the photoelectric current I is also inversely proportional to the square of the distance:

$$I \propto \frac{1}{r^2} \implies I_1 r_1^2 = I_2 r_2^2$$

We can solve for the new current I_2 :

$$I_2 = I_1 \left(\frac{r_1}{r_2} \right)^2$$

Substitute the given values $I_1 = 7.50 \mu\text{A}$, $r_1 = 3.0 \text{ m}$, and $r_2 = 5.0 \text{ m}$:

$$I_2 = 7.50 \left(\frac{3.0}{5.0} \right)^2 = 7.50 \left(\frac{3}{5} \right)^2 = 7.50 \left(\frac{9}{25} \right)$$

Calculate the final value:

$$I_2 = 7.50 \times 0.36 = 2.70 \mu\text{A}$$

The new photoelectric current is $2.70 \mu\text{A}$.